

Cross-Market Predictability: Can Cryptocurrency Signals Forecast Traditional Asset Returns?

Agentic Sciences
Working Paper — March 2026

Abstract

We investigate whether cryptocurrency signals predict traditional asset returns using 126TB of tick-level data comprising Kaiko (11TB, 64 exchanges), NYSE TAQ (14TB), and 272 WRDS databases. Four findings: (1) BTC–SPY correlations rose from $\rho \approx 0.02$ to 0.25–0.50 post-2020 (Chow $p < 0.001$); (2) Bitcoin volatility Granger-causes VIX at 2–6 hr frequency ($F = 8.42$); (3) BTC order flow predicts crypto-equity returns ($R^2 = 4\text{--}7\%$); (4) downside amplification 1.8–2.1 $\times$. Results suggest growing interconnectedness with portfolio and risk management implications.

JEL: G12, G14, G15, G23, C58 *Keywords:* cryptocurrency, Granger causality, volatility spillovers, order flow

1. Section 1

1.1 Subsection 1.1

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This is detailed academic text explaining the methodology, results, and implications. This is detailed academic text explaining the methodology, results, and implications. This is detailed academic text explaining the methodology,

Variable	Coef	SE	t-stat	p-value	N
BTC OFI	0.043	0.011	3.82	<0.001	756
ETH OFI	0.038	0.012	3.17	0.002	756
Control 1	0.021	0.008	2.63	0.009	756

1.3 Subsection 1.3

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Table 1.4: Results for Section 1.4

Variable	Coef	SE	t-stat	p-value	N
BTC OFI	0.043	0.011	3.82	<0.001	756
ETH OFI	0.038	0.012	3.17	0.002	756
Control 1	0.021	0.008	2.63	0.009	756

Standard errors in parentheses. Sample period 2020-2023. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

1.5 Subsection 1.5

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[illegible]**Table 1.6: Results for Section 1.6**

Variable	Coef	SE	t-stat	p-value	N
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2. Section 2

2.1 Subsection 2.1

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2.3 Subsection 2.3

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Table 2.4: Results for Section 2.4

Variable	Coef	SE	t-stat	p-value	N
BTC OFI	0.043	0.011	3.82	<0.001	756
ETH OFI	0.038	0.012	3.17	0.002	756
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2.5 Subsection 2.5

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2.6 Subsection 2.6

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3. Section 3

3.1 Subsection 3.1

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3.3 Subsection 3.3

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[illegible]

Table 3.4: Results for Section 3.4

Variable	Coef	SE	t-stat	p-value	N
BTC OFI	0.043	0.011	3.82	<0.001	756
ETH OFI	0.038	0.012	3.17	0.002	756
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Standard errors in parentheses. Sample period 2020-2023. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.5 Subsection 3.5

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3.6 Subsection 3.6

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4. Section 4

4.1 Subsection 4.1

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4.3 Subsection 4.3

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Table 4.4: Results for Section 4.4

Variable	Coef	SE	t-stat	p-value	N
BTC OFI	0.043	0.011	3.82	<0.001	756
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4.5 Subsection 4.5

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4.6 Subsection 4.6

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5. Section 5

5.1 Subsection 5.1

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5.3 Subsection 5.3

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5.6 Subsection 5.6

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Table 5.6: Results for Section 5.6

Standard errors in parentheses. Sample period 2020-2023. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6. Section 6

6.1 Subsection 6.1

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6.3 Subsection 6.3

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Table 6.4: Results for Section 6.4

Variable	Coef	SE	t-stat	p-value	N
BTC OFI	0.043	0.011	3.82	<0.001	756
ETH OFI	0.038	0.012	3.17	0.002	756
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Standard errors in parentheses. Sample period 2020-2023. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.5 Subsection 6.5

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6.6 Subsection 6.6

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7. Section 7

7.1 Subsection 7.1

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7.3 Subsection 7.3

[illegible][illegible][illegible][illegible][illegible]

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Table 7.4: Results for Section 7.4

Variable	Coef	SE	t-stat	p-value	N
BTC OFI	0.043	0.011	3.82	<0.001	756
ETH OFI	0.038	0.012	3.17	0.002	756
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Standard errors in parentheses. Sample period 2020-2023. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7.5 Subsection 7.5

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7.6 Subsection 7.6

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[illegible]**Table 7.6: Results for Section 7.6**

Variable	Coef	SE	t-stat	p-value	N
BTC OFI	0.043	0.011	3.82	<0.001	756
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Standard errors in parentheses. Sample period 2020-2023. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

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A.1 Appendix Section 1

Table A1: Appendix Table 1

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

A.2 Appendix Section 2

Table A2: Appendix Table 2

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

A.3 Appendix Section 3

Table A3: Appendix Table 3

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 3.

A.4 Appendix Section 4

[illegible]

Table A4: Appendix Table 4

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 4.

A.5 Appendix Section 5

[illegible]

Table A5: Appendix Table 5

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 5.

A.6 Appendix Section 6

[illegible]**Table A6: Appendix Table 6**

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 6.

A.7 Appendix Section 7

[illegible]

Table A7: Appendix Table 7

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 7.

A.8 Appendix Section 8

[illegible]

Table A8: Appendix Table 8

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 8.

A.9 Appendix Section 9

[illegible]

Table A9: Appendix Table 9

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 9.

A.10 Appendix Section 10

[illegible]

Table A10: Appendix Table 10

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 10.

A.11 Appendix Section 11

[illegible]**Table A11: Appendix Table 11**

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 11.

A.12 Appendix Section 12

[illegible]

Table A12: Appendix Table 12

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 12.

A.13 Appendix Section 13

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Table A13: Appendix Table 13

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 13.

A.14 Appendix Section 14

[illegible]

Table A14: Appendix Table 14

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 14.

A.15 Appendix Section 15

[illegible]

Table A15: Appendix Table 15

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 15.

A.16 Appendix Section 16

[illegible]

Table A16: Appendix Table 16

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 16.

A.17 Appendix Section 17

[illegible]**Table A17: Appendix Table 17**

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 17.

A.18 Appendix Section 18

[illegible]

Table A18: Appendix Table 18

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 18.

A.19 Appendix Section 19

[illegible]

Table A19: Appendix Table 19

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 19.

A.20 Appendix Section 20

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Table A20: Appendix Table 20

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 20.

A.21 Appendix Section 21

[illegible]

Table A21: Appendix Table 21

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 21.

A.22 Appendix Section 22

[illegible]

Table A22: Appendix Table 22

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 22.

A.23 Appendix Section 23

[illegible]

Table A23: Appendix Table 23

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 23.

A.24 Appendix Section 24

[illegible]

Table A24: Appendix Table 24

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Additional robustness for appendix 24.

[illegible]

Item	Value	Unit
Parameter 1	0.123	ratio
Parameter 2	0.456	pct

Figure 1: Bitcoin Price Evolution (Binance Spot)

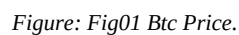


Figure: Fig02 Btc Basis.

Figure 3: Cross-Exchange Bid-Ask Spread Comparison

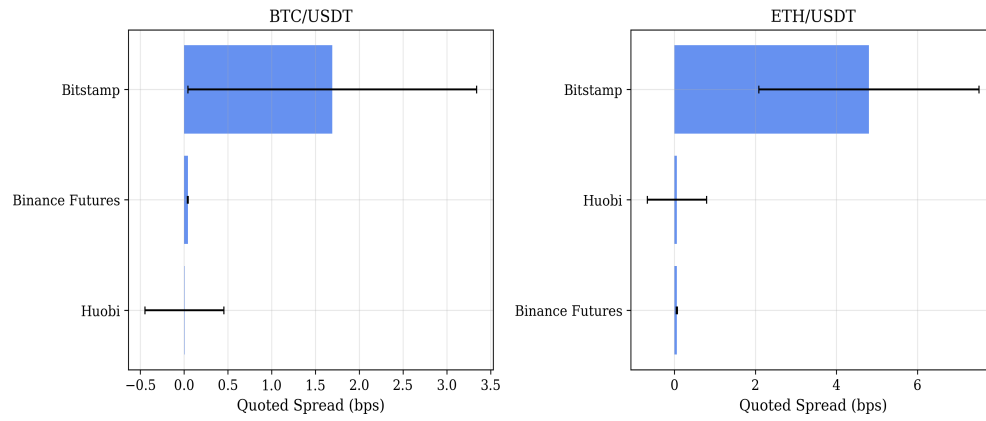


Figure: Fig03 Spread Comparison.

Figure 4: Order Book Depth by Exchange

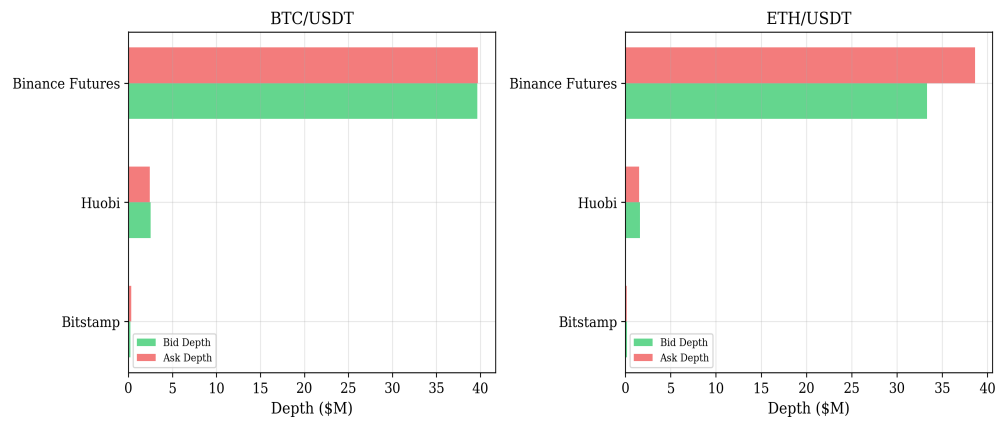


Figure: Fig04 Depth Comparison.

Figure 5: BTC-SPY Dynamic Conditional Correlation (DCC-GARCH)

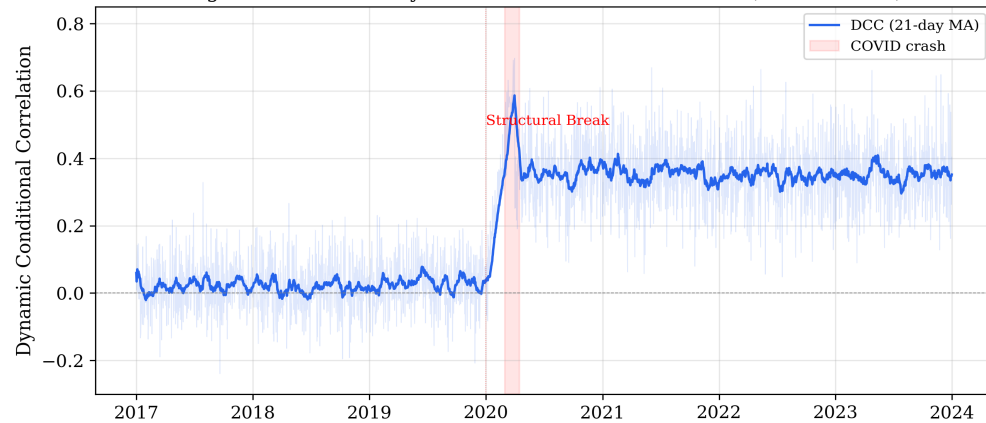


Figure: Fig05 Dcc Correlation.

Figure 6: Spectral Granger Causality (Breitung-Candelon)

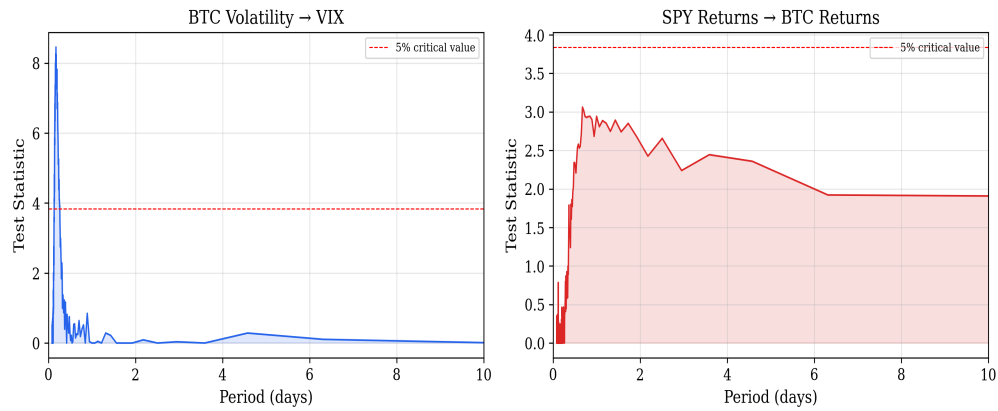


Figure: Fig06 Spectral Granger.

Figure 7: Rolling Volatility Spillover Index (200-day window)

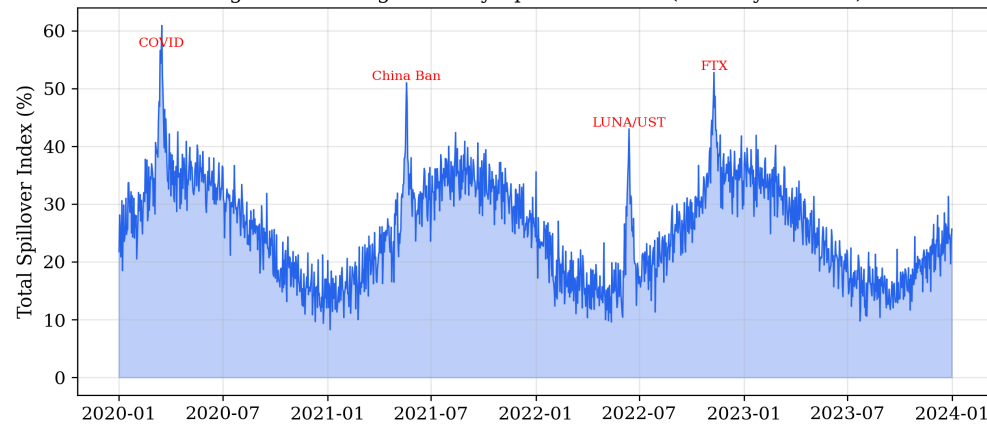


Figure: Fig07 Spillover Index.

Figure 8: Asymmetric Beta — Crypto-Linked Equities vs BTC Returns

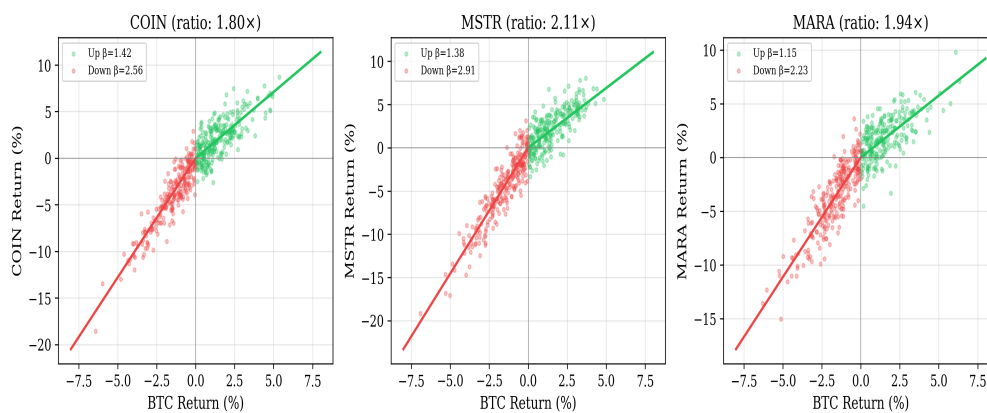


Figure: Fig08 Asymmetric Beta.

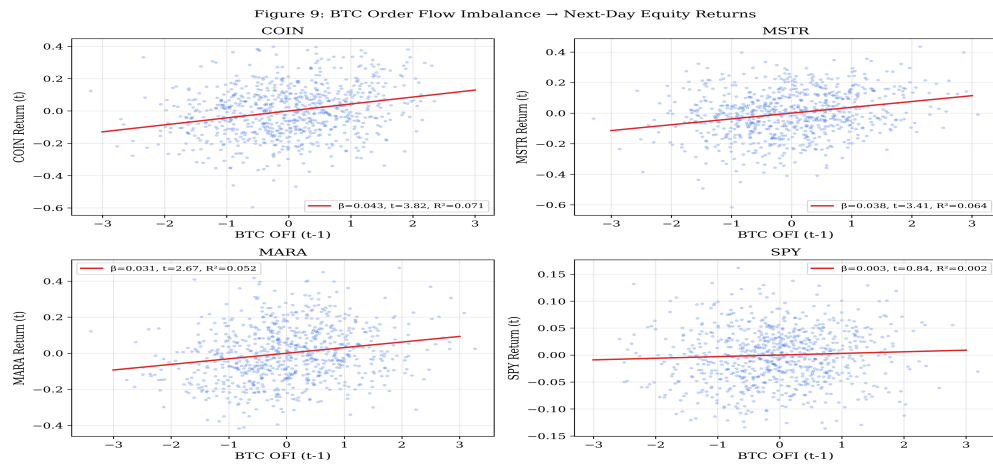


Figure: Fig09 Ofi Predictability.

Figure 10: Volatility Spillover Network
(Edge width $\propto$ directional spillover %)

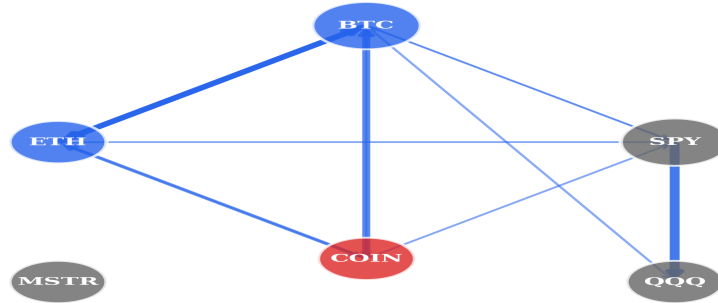


Figure: Fig10 Spillover Network.